

HORIZON — Business Model Canvas

Dependency-Aware Autonomous Infrastructure Recovery Platform

KEY PARTNERS	KEY ACTIVITIES	VALUE PROPOSITIONS	CUSTOMER RELATIONSHIPS	CUSTOMER SEGMENTS
Potential integrations <i>(not confirmed partnerships)</i> - Cloud providers (AWS/GCP India, E2E, Yotta) — targets & marketplace - Observability (Datadog, Dynatrace, Prometheus) — failure signals - Indian LLM vendors (e.g. Sarvam) — optional NLU assist - EVM testnet operators — optional audit anchors (MST used in prototype) - MSPs / SRE consultancies — implementation channel - Incubators (T-Hub, NASSCOM DeepTech) — GTM support	Core (near-term) - Harden DAG engine, BFS blast radius, topo recovery planner - Health probes; simulated → Docker/K8s connectors - Risk classification & approval-gate UX - Audit trail integrity & export - Failure-injection evaluation (MTTR, order violations) Commercial - Demo sandbox & self-serve onboarding - Technical content & outage case studies - Enterprise POC design	Primary outcomes (proposed) - Correct-order recovery: restore only after dependencies healthy - Blast-radius visibility on live topology at failure time - Low-risk auto-execute; high-risk pause for human (or optional crypto) approval - Health verification between recovery steps - Traceable recovery timeline & audit record Secondary (roadmap) - Optional LLM diagnostic suggestions (non-critical path) - Multilingual operator assist (proposed) - On-chain audit anchors (testnet prototype) <i>MTTR % claims not statistically validated.</i>	Self-serve / PLG - Interactive sandbox & DAG visualizer - Docs + MCP/CLI entry (roadmap) Collaborative - Shared incident view; Slack/Teams (planned) Enterprise - Guided onboarding & custom playbooks - TAM support for higher tiers - Chaos-drill validation support	Initial focus - Cloud-native Indian startups & scale-ups (lean SRE, multi-service K8s) Pain: out-of-order restarts, war-room lag Buyers: CTO / VP Eng / Head of SRE Near-term expansion - Regulated BFSI / FinTech (audit & control gates) - HealthTech / platforms needing resilience evidence Users: SRE, DevOps, Incident Commanders Future: MSPs, multi-region enterprises

KEY RESOURCES	CHANNELS
Existing - TypeScript/React monorepo + pure in-browser recovery engine - DependencyGraph (adjacency, cycle detect, topo order, BFS blast radius) - Simulated 7-node cluster & recovery timeline UI - Optional FastAPI, Firebase RTDB, MST testnet contracts - Research paper + public GitHub (Precise-Goals/horizon) + live demo (horizon-aiops.vercel.app) To build / acquire - Production connectors · formal evaluation data · security audits · counsel-reviewed IP	Available now - Live demo & visual DAG explorer - GitHub + technical documentation - Hackathon / mentor presentations Planned - Product-led growth (sandbox → paid) - Developer MCP / CLI directory - Direct enterprise outreach - Cloud marketplace listings · Partner MSP referrals

COST STRUCTURE	REVENUE STREAMS
Fixed / semi-fixed - Core engineering (distributed systems, frontend, connectors) - Cloud hosting for control plane (demo on Vercel + Firebase) - Security testing; smart-contract audits if Web3 path retained - Legal / provisional patent counsel (India CRI framing) Variable / usage-linked - Customer probe/execution agents · optional LLM tokens · batched chain gas · enterprise support <i>Margin % and per-incident costs in earlier drafts are not validated.</i>	Proposed SaaS tiers (indicative, not validated pricing) - Explorer — Free: limited nodes, visual DAG, basic audit - Guardian — ~₹4,999/mo: auto low-risk recovery, approval gates, ~50 nodes - Sentinel — ~₹24,999/mo: higher scale, agent assist, MCP access, export - Enterprise — Custom: dedicated/on-prem options, SLA, unlimited nodes Future levers - Usage metering · professional services · playbook marketplace commission <i>No paying customers or ARR claimed at this stage.</i>

Section B — Business Model Explanation

How Horizon creates value

Horizon addresses a concrete operational failure mode: restoring interdependent services out of dependency order. When a database is down, restarting dependent APIs produces crash loops and extends outage duration. The platform maintains a live directed dependency graph, computes blast radius on failure, generates a topological recovery sequence, executes low-risk steps automatically, and pauses for human (or optional cryptographic) approval on high-risk steps. Each recovered node is health-checked before dependents proceed, and transitions are recorded in an audit trail.

Value is therefore expressed as fewer dependency-order violations, clearer blast-radius visibility, supervised automation, and a recoverable incident timeline—not as generic “AIOps” or unvalidated MTTR percentages.

How value is delivered

Delivery today is via a deployable prototype: React/Vite web UI (Command Center, interactive DAG, recovery timeline, audit view), pure TypeScript recovery engine in-browser, optional FastAPI services, Firebase real-time state, and simulated seven-node cluster (PostgreSQL, Redis, Auth, Gateway, Frontend, Payment, etc.). Recovery actions are simulated or target local Docker/Kubernetes where configured. The critical path (graph construction, topological order, blast-radius BFS, approval gating) is deterministic and does not require LLM calls.

Go-to-market delivery starts with a self-serve sandbox and live demonstration, then expands to guided enterprise pilots and eventual connector-based deployment into customer environments.

How revenue is captured

The proposed model is tiered SaaS (free Explorer for evaluation; paid Guardian/Sentinel by node scale and feature depth; custom Enterprise). Optional future levers include usage metering, professional services, and a recovery-playbook marketplace. Pricing figures (₹4,999 / ₹24,999) are strategic proposals only; no willingness-to-pay studies or contracts exist yet. Web3 NFT subscription gating is implemented on testnet as a prototype path and remains optional for core product value.

Why the model is commercially plausible

The problem is real and expensive for teams running multi-tier microservices. Commercial AIOps and orchestration tools (StackStorm, Rundeck, Kubernetes controllers, Datadog, ServiceNow, Rubrik ABR) already show demand for sequenced or policy-gated recovery. Horizon’s coherent closed loop—live topology → dynamic sequence → risk gates → health verification → audit—maps cleanly onto that demand, even if individual mechanisms are known. A free sandbox lowers adoption friction; regulated buyers may pay for control and auditability.

Plausibility is constrained by prototype maturity, strong prior art (notably US12556441B1), and the need for measured technical effect before enterprise claims.

Assumptions that still require validation

See Section D table. Critical items: measurable MTTR and order-violation improvement versus baselines; willingness to pay at proposed price points; production connector reliability; residual patentability of narrow closed-loop claims; and whether blockchain audit anchoring adds commercial value beyond standard immutable logs.

Section C — Business Model Priorities

Primary initial customer segment

Cloud-native Indian startups and scale-ups (Series A–C) operating 20–200+ microservices on Kubernetes with lean SRE teams. Pain is acute, decision cycles are shorter than large regulated enterprises, and a visual sandbox can convert interest into a pilot.

Strongest proposed value proposition

Dependency-constrained recovery with mandatory approval for high-risk steps and health verification between steps—i.e., “recover in the right order, under human control, with a full trail.” This is more defensible and useful than broad “autonomous AI recovery” claims.

Recommended initial pricing model

Free Explorer (evaluation + limited nodes) + single mid-tier paid plan (Guardian-class) focused on auto low-risk recovery and approval gates. Defer multi-tier complexity and NFT gating until product-market fit signals appear. Treat all rupee amounts as proposals pending customer discovery.

Most practical customer acquisition channel

Product-led demonstration: live Command Center + DAG explorer + recorded failure/recovery walkthroughs, amplified through GitHub, technical blogs on known outages, and hackathon/mentor networks. Parallel light outbound to SRE leaders at target startups.

Essential integrations and partnerships

Kubernetes/Docker execution adapters; at least one observability signal source (Prometheus or a commercial APM webhook); Slack/Teams notification. Treat Sarvam, MST, and cloud marketplaces as optional or later-stage opportunities until formal agreements exist.

Three most important commercial risks

1. Prior-art collision — broad dependency-mapped recovery claims face US12556441B1 and related art; residual novelty is narrow and unproven.
2. Prototype-to-production gap — simulated cluster success does not equal safe execution on customer estates; connector quality and fail-safe behavior must be proven.
3. Willingness to pay — SREs may value the demo yet still rely on existing runbooks/orchestrators unless measurable MTTR or risk-reduction gains are shown.

Three immediate validation priorities

1. Controlled evaluation against baselines (manual, static playbook, Horizon with sequencing off, full Horizon) measuring MTTR, dependency-order violations, and success rate.
2. Customer discovery interviews with 15–20 target SRE/CTO personas on pain severity, current tools, and price sensitivity.
3. Counsel opinion on residual novelty for narrow closed-loop claims (mid-recovery re-planning, dual-path approval, state machine) before any provisional filing.

Section D — Assumptions and Validation

Assumption	Current Evidence	Confidence	Validation Required
Dependency-aware sequencing reduces order-violation errors vs static runbooks	Engine implements topo sort + health gates; paper notes only preliminary simulation observations	Medium	Controlled A/B with baselines; publish sample size, variance, success rate
Customers will pay ~₹5k–25k/mo for gated autonomous recovery	No pricing tests or contracts; competitive tools exist at different price points	Low	Discovery interviews + pilot pricing experiments
Live topology + closed-loop control is commercially differentiated	US12556441B1 and Rubrik ABR / AUTO-OPS closely related; paper rates novelty moderate at best	Low–Medium	Counsel prior-art opinion; focus claims on mid-recovery re-plan + dual approval path
Blockchain audit anchors are valued by buyers	Testnet contracts exist; standard append-only logs may suffice for most	Low	Ask regulated buyers whether on-chain proof is required vs exportable signed logs
Sarvam / MST are strategic partners	Integrations referenced in drafts; no public partnership announcements found	Low	Formal agreements or re-label as vendor integrations
India AIOps demand supports rapid SaaS growth	Third-party reports show growth in India AIOps-related markets (high CAGR); not Horizon-specific	Medium	Segment-specific TAM for recovery orchestration, not full AIOps
Prototype can reach production safety	Cycle detection & high-risk gates present; no auto-rollback, limited multi-failure stress	Medium	Failure-mode testing (stale graph, partial failure, approval timeout) + security review

Implementation reality check (summary)

Verified in repository and paper: DependencyGraph with cycle detection, topological recovery order, blast-radius BFS, risk-classified steps, approval pause, health-check gate, audit logging, simulated multi-tier demo, React Command Center, optional MST testnet contracts. Research paper rates overall technical differentiation ~6/10 and novelty potential for narrow claims ~4–5/10, with high patentability risk for broad claims.

Not verified as production-ready: multi-region recovery, formal model-checking of plans, automatic rollback of partial sequences, statistically significant MTTR gains, confirmed commercial partnerships, or regulatory certifications (CERT-In / DPDP / SOC2).

Recommendation for hackathon and early commercialization: lead with the live correct-order recovery demonstration, visible approval gate, blast-radius visualization, and audit timeline. Treat patent language and aggressive financial projections as secondary until evidence strengthens.

— End of Business Model Canvas Deliverable —